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USER BOOK TITLE: Water architecture in South Asia : a study of types, development, and meanings /

CHAPTER TITLE: Wells

BOOK AUTHOR: Hegewald, Julia A. B.

EDITION:

VOLUME:

PUBLISHER: Brill,

YEAR: 2002

PAGES: 148-177

ISBN: 9789004123526

LCCN:

OCLC #: 48621347

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LCCN:
OCLC #: 48621347
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WATER ARCHITECTURE IN SOUTH ASIA

A Study of Types, Development and Meanings

BY

JULIA A.B. HEGEWALD



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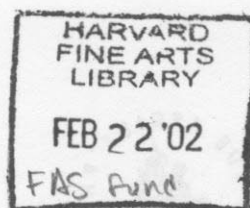
Water architecture in South Asia : a study of types, development and meanings / by Julia A.B. Hegewald. - Leiden ; Boston ; Köln : Brill, 2001

(Studies in Asian art and archaeology ; Vol. 24)

ISBN 90-04-12352-0

Library of Congress Cataloging-in-Publication Data

Library of Congress Cataloging-in-Publication Data is also available



ISSN 1380-782X
ISBN 90 04 12352 0

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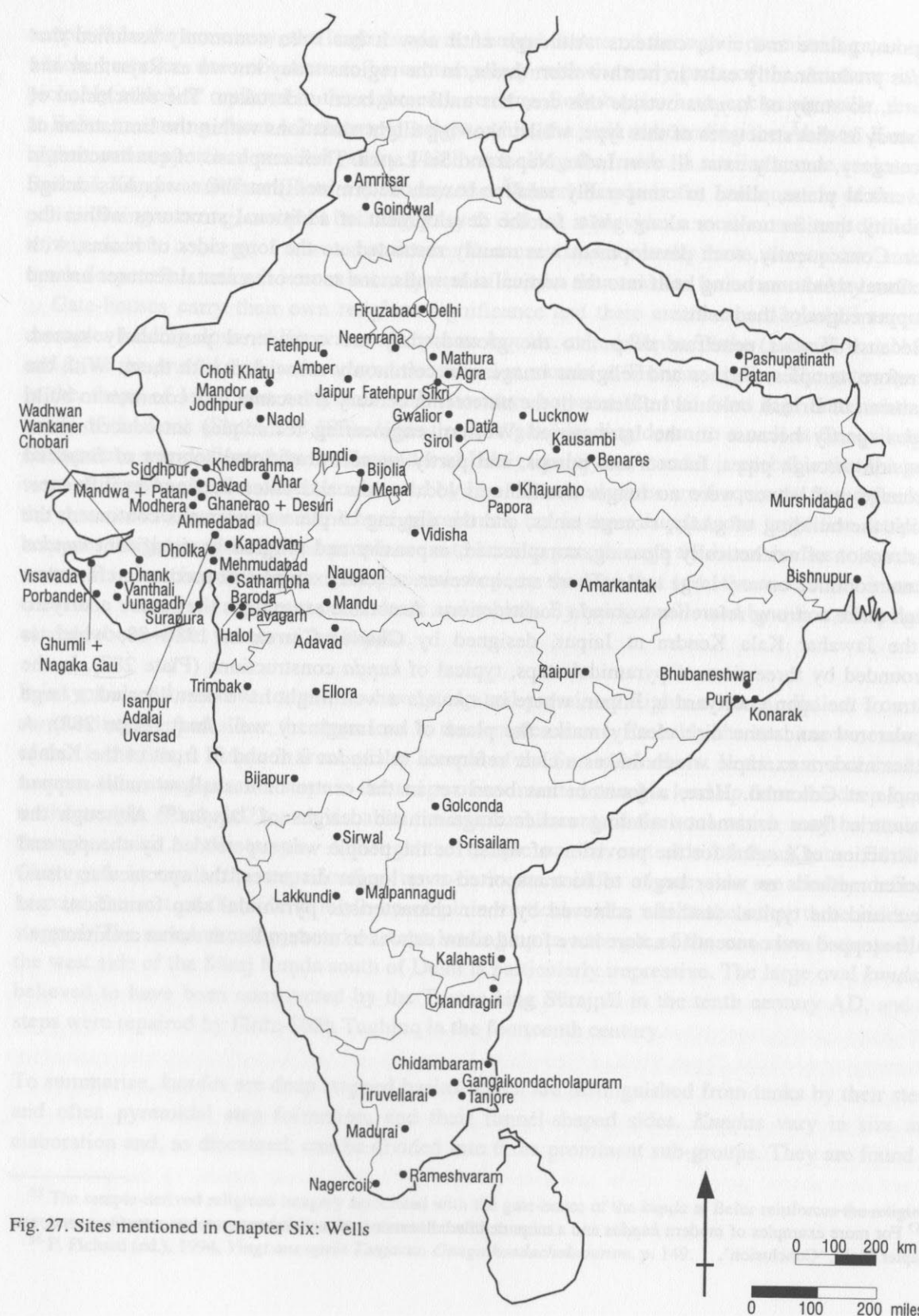


Fig. 27. Sites mentioned in Chapter Six: Wells

CHAPTER SIX

WELLS

1. INTRODUCTION

Wells are vertical cylindrical shafts which penetrate deep into the ground and always tap ground water,¹ the most reliable source of water provision. Among the names most frequently used to refer to wells are *kūpa* (Sanskrit), *kuām* (Hindi), *kūo* (Gujarati) and *ināra* (Nepali). The depth of the well shaft depends on that of the water table, which can at times lie as much as 100 to 200 metres below the surface. Most wells are circular, but square, octagonal and hexagonal shafts exist, which are typical of more elaborate structures where the well shaft represents only one component of a larger well monument. At the top, the wells are usually surrounded by a parapet wall, and in an ordinary draw well, this low wall is the only part of the water structure which is visible over ground. At the well-head, a permanent pulley mechanism is frequently installed to haul up buckets or leather bags (*pūrs*); this may be done manually or by bullocks turning a Persian wheel. Nowadays, electric pumps also are fitted. Among the various types of water architecture, wells expose the smallest surface area to the sun; even during the dry season, they keep water longer and fresher than most other water structures. Particularly in desert areas, where rain falls only during a few months of the year, they are vital for the survival of men, animals and crops. Wells are found all over South Asia and coexist alongside *ghāṭs*, tanks and *kuṇḍas*.

The earliest known wells in South Asia have been found in the excavations at Mohenjo-daro and Harappa in the area of the modern state of Pakistan. They are deep brick built circular structures and date from about 2800 BC. Many other wells have early dates, but because in a plain draw well there is not much space for architectural display and decoration, and the parapet walls are the most likely part to have been rebuilt, exact dating of wells is often difficult. During the early periods, wells were frequently constructed from prefabricated circular rings made of stone or burned brick. Such wells are commonly referred to as ring-wells.²

Wells are usually located in public places which are easily accessible to the people;³ larger houses and palaces have private wells. Public wells are to be found in the streets and on the squares of villages and towns, where they provide water for everyday domestic needs. Frequently, they have a raised stone platform surrounding the well shaft which is used for the washing of clothes and other items at the well (Fig. 28). They may also be located at city gates, at cross-roads

¹ There are very few exceptions, such as the Champā Bāvlī at Mandu and the 'well' in the Bolanāth Mandir at Bijolia, which receive water from nearby water sources through subterranean channels.

² One such ring-well, made of terracotta, survives near the Council Chamber at Polonnaruva, and two granite ring-wells may be seen in the compound of the Naṭarāja Temple at Chidambaram.

³ Both the *Aparājitaṭṭṭchā* and the *Mānasāra* state that wells should be located in places where "they can conveniently be reached by a large number of inhabitants" (L. M. Dubey, 1987, *Aparājitaṭṭṭchā—A Critical Study*, p. 120).

and on overland roads to provide water to travellers. Many wells have stone troughs beside them to water animals. Situated in or near the fields, they are also used for irrigation. Wells are meeting places for the community, and believed to be sacred, turning many of them into active places of worship.

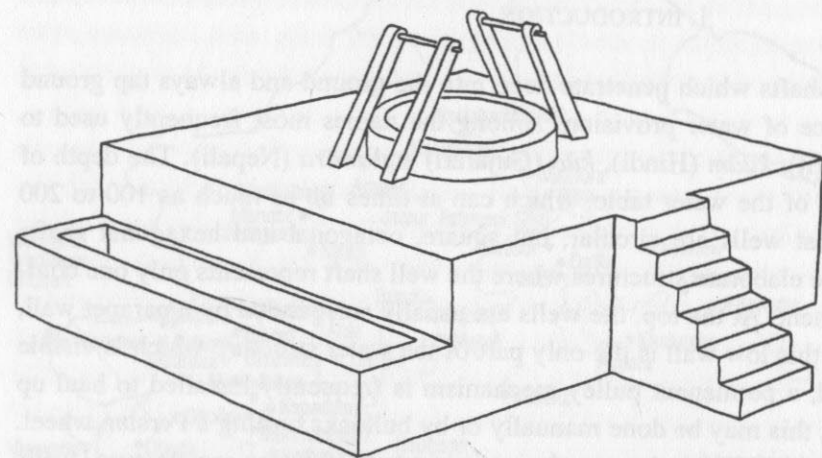


Fig. 28. Draw well in a raised platform

Wells are frequently found in connection with Hindu, Buddhist and Jaina temples and also with mosques and gurdwaras. They can either be located directly adjacent to religious monuments, or further away in their compounds. In a few cases they are even found within a religious edifice itself.⁴ Wells have strong religious connotations because they penetrate deep into the ground, which is regarded as the abode of gods and spirits. On the one hand, this may have beneficent effects on the people using the well, because the ground water is believed to be imbued with divine powers, to cleanse body and soul, to be regenerating, healing and fertilising, and to enhance prosperity. On the other hand, wells may be haunted by malevolent powers, particularly by the restless souls of people who died a violent death or for whom no proper death rituals were conducted by their families.⁵ Such evil spirits (*bhūts*) may inflict disease and disaster on people approaching the wells. The close relationship between death and wells is apparent from the many *satī*- and hero-stones found next to well monuments, as for example the hero-stones on the side of the Jñān Vāv at Visavada near Porbander in the south-west of Gujarat, or the Muslim grave adjacent to the Bīja Maṇḍal Bāvlī at Vidisha. Wells were frequently constructed to commemorate the dead, as for example the Rāṇī Vāv⁶ at Patan, built by Queen Udayamati in memory of her

⁴ See the discussion of 'well-temples' later on in this chapter.

⁵ A. W. Entwistle, 1987, *Braj: Centre of Krishna Pilgrimage*, p. 295.

⁶ Various spellings exist for the name of this well. 'Rāṇī Vāv' is the Sanskrit, 'Rānī-kī Vāv' the Hindi and 'Rāṇī Vāv' the Gujarati version of the name, while K. Mankodi, in his monograph on the well (1991, *The Queen's Stepwell at Patan*), calls it the 'Rāṇkī Vāv'.

deceased husband Bhīmadeva I.⁷ In order to please and retain the gods, and to propitiate the evil powers who may be present in wells, religious images are frequently installed⁸ and religious rites conducted at wells. The Nāga Kūpa at Benares is believed to be inhabited by a dangerous snake, a *nāga-rāja*, who is propitiated by offerings of milk. Many wells are closely connected with the cult of Jaladevī, the goddess of water, or with mother goddesses, which is evident from their names. Examples are the Aṅkol Mātā Stepwell at Davad (Dāvad) and the Āśāpurī Vāv at Ahmedabad, both in Gujarat: *mātā* means mother, and Āśāpurī is the name of a mother goddess.⁹ It is interesting to observe that *kuṇḍas*, most of which also contain a well, are more typically associated with the male god Śiva, and some, such as the *kuṇḍa*-well hybrid adjacent to the Kuśavart Tīrtha at Trimbak, take the shape of an inverted *linga*, while the narrow vertical shafts of the wells are predominantly connected with mother earth and female goddesses.¹⁰

There is a multitude of myths associated with well monuments, many of which further emphasise the religious and fertility aspects of such constructions. These myths are given tangible form in the local customs, rites and religious ceremonies that have developed around the wells. The Saptasamudrikā Kūpa (Well of the Seven Seas) in the compound of the Museum at Mathura, for example, is traditionally visited by newly-married women,¹¹ and in other parts of India, young brides and their husbands bathe in a well immediately after their wedding to become imbued with fertility and health. A bath in the Marīci Kuṇḍa Well¹² adjacent to the Mukteśvara Temple at Bhuvaneshwar is believed to make women fertile who cannot have children, and the tossing of coins into the same well is meant to generate prosperity. In Braj, newly born babies are taken to a well ten days after their birth to be sprinkled with sacred well water, and offerings are made to the well as they would be to a god in a temple.¹³ Young mothers pour cow's milk on the parapet of wells to secure a plentiful supply of milk to their children, and young girls believe that pots of sprouting grain placed in a stepwell will bring good luck to their marriage.¹⁴ Well water is also closely related to *amṛta*, and in Muslim areas the village well is frequently addressed as *Khawāja Khizr*.¹⁵ He and Alexander the Great searched for the secret location of the elixir of life, but only *Khawāja Khizr* is believed to have found it and gained immortality. Frequently, great wisdom and clairvoyance is attributed to wells in local myths. A well on Rājarājeśvarī Ghāt at Pashupatinath, for example, had to be closed because it mirrored the future too faithfully. People who look into the

⁷ The spelling of the name Udayamati in the primary sources is not consistent while most resort to 'Udayamati', others spell her name 'Udayamatī'.

⁸ Images installed on the inside or in close connection with well shafts will be discussed in the section 'From Wells and Temples to Well-Temples', later on in the chapter.

⁹ N. Gutschow & J. Pieper, 1978, *Indien: Von den Klöstern des Himalaya zu den Tempelstädten Südindiens*, p. 317; J. Jain-Neubauer, 1981, *The Stepwells of Gujarat in Art-Historical Perspective*, p. 5.

¹⁰ A. W. Entwistle in his study of Braj (1987, p. 295), however, describes how some of the wells of that region are associated with Varuṇa, the Vedic god of the water, whom the local people refer to as 'Kūāmvālā', the 'well man'.

¹¹ *Ibid.*, p. 296.

¹² In Orissa, 'Mārīci' is frequently spelled 'Marīci' (See also W. Smith, 1994, *The Mukteśvara Temple in Bhuvaneshwar*, pp. 34, 42).

¹³ Entwistle, 1987, p. 295.

¹⁴ Livingston, 1995, 'The Stepwells and Stepped-Ponds of Western India', p. 18.

¹⁵ Rama (no initial), 1995, *Our Water Resources*, p. 20.

Kāl Kūpa (Well of Fate), situated near the Bhaironāth Temple at Benares, at noon and do not see their own shadow on the water are believed to be doomed to die within the next six months, unless they gain the help of Mahākāl or Śiva to intervene with Yama, the god of death, on their behalf.¹⁶

2. VARIATIONS IN WELL DESIGN

Draw Wells

The simplest version of a well is an ordinary draw well consisting in a vertical well shaft penetrating deep into the earth to reach the ground water table, and in most cases a pulley mechanism. In cases where no mechanism exists, women bring their own bucket on a string to raise the water. The shafts of simple draw wells are usually circular or square. Wells which have shafts of only about one to two metres in diameter usually have parapet walls which are about one metre high, a comfortable height for a person to lean against and pull up water.¹⁷ Such wells may be seen in the towns of the Kathmandu Valley, where they are located in most squares. Wells still possessing original or at least relatively old parapet walls often have beautiful carvings displaying water-related divinities and plant ornamentation.¹⁸

Wells with such high parapet walls often have a column, made of stone, wood or iron, on either side of the well, upon which rests a beam. From the under surface of the beam is suspended a wheel or other machinery to haul up water. Examples may be seen in the outer enclosure of the Natarāja Temple complex at Chidambaram, and on the south side of the Vaḍa-Kailāsanātha Temple in the compound of the Bṛhadiśvara Temple at Gangaikondacholapuram.¹⁹ It is typical of southern Maharashtra, for example the region around Adavad, to have square wells with two walls built on opposite sides of the shaft which are stepped and rise towards one side. Wooden poles, over which runs the rope to raise a bucket, can be placed over the well shaft on different levels of the stepped sides to adapt to the person working the pulley (Plate 201). In other cases, two metal staffs with a further one fixed crosswise which runs through a small wheel over which the rope is pulled, reach out from the parapet wall into the centre of the well. These metal constructions seem to be later replacements of stone brackets which had a strong wooden rod fixed crosswise, as can still be seen in the red sandstone well in the enclosure of the Tāj Mahal at Agra. Wider wells, with a diameter of about two to three metres, often have very low or no parapet walls at all (Plate 200). Such examples frequently have narrow platforms running around the

¹⁶ E. B. Havell, 1905, *Benares the Sacred City: Sketches of Hindu Life and Religion*, p. 185.

¹⁷ An unusual example, where well water is not pulled up from the side of the parapet wall, is the Āgam Kuāṃ at Patna. At this well, a thin stone slab, wide enough to stand on it, is fixed over the opening of the well, and the women stand on the slab above the shaft and pull up the buckets from below.

¹⁸ A draw well in the north of Patan in Nepal, for instance, has an image of Viṣṇu Nārāyaṇa resting on the cosmic snake Śeṣa or Ananta carved on the outer rim of the parapet wall.

¹⁹ There are also a few examples of wells flanked by such columns which, however, have no parapet walls surrounding the well shafts. One such example is to be found in front of the mosque in the complex of the Baṛā Imāmbārā at Lucknow, which is used for ablutions before prayer. The well shaft is contained in a very low platform, about ten centimetres in height, which also supports the two carved sandstone pillars.

well shaft on the inside, supported underneath by beautifully carved brackets. Examples of this are the Bābūke Vāv near Ghumli and a draw well in the village of Amber.

Such wide wells are often located in the middle of raised platforms which provide a firm and clean surface, raised above the dust of the ground, to wash clothing or produce. The platforms also prevent people or animals from falling into the wells which, without a parapet wall proper, would just be a hole in the ground. Wells with raised platforms can be found in towns, as for example in Bundi, or in the countryside, as in the *dharamśālās* on the Pañcakroṣī Road around Benares or on the country road from Datia to Sirol village in Madhya Pradesh. The latter also has a stone trough adjacent to its well shaft which can be filled to enable animals to drink, a feature typical of wells located in villages or fields. Half way along the trough is a small shrine with a domed roof. Further examples of this, and the religious connotations of wells in general will be explored later on in this chapter. Troughs and channels are used to direct water away from the well to irrigate the fields around. The Bābūke Vāv near Ghumli has both a cattle trough and an irrigation channel. Because a lot of water is needed for the irrigation of fields, wells which are used for agricultural purposes often have a Persian wheel which can be turned by bullocks to pump up water (Plate 202). Another way of lifting water with the help of oxen is to construct ramps leading down and away from the well shafts. By walking up and down such ramps, the animals lift the water in large leather bags.

The Shekavati region of Rajasthan is especially well known for a certain kind of draw well which is usually raised in a high terrace (often up to three metres high) and flanked by two, or more often by four, tall slender towers which resemble minaret-like structures (Plate 26). The flanking towers, mark the wells and make them more easily visible to travellers in this dry desert area. These wells, too, connect with cattle ramps, troughs and irrigation channels. Although the association of wells in terraces surrounded by tall towers is very common in the Shekavati area, this feature is not unique to this region of Rajasthan. The towers in this region are particularly tall, but slightly shorter examples arranged in the same way exist elsewhere. A well very similar to many structures in Fatehpur, for instance, is to be found in the later palace on the edge of Jaisalmer, and the areas around Kausambi (Kauśāmbī) in Uttar Pradesh, Khajuraho and Gwalior, are literally studded with wells in the 'Shekavati style' (Plate 203).²⁰

Ordinary draw wells occupy comparatively little space, consequently they are more frequently found in confined temple compounds, than larger linear stepwells. Every temple needs water in its vicinity for the bathing of the image housed in the shrine room, for the execution of temple rituals, and for the ablutions of priests and worshippers. Although many temples have small ablution basins or proper bathing tanks,²¹ the water for temple rituals is wherever possible, taken from a sacred well in the temple's vicinity. Examples of draw wells in temple compounds may be

²⁰ Another feature, frequently connected with this kind of well monument, is the presence of thin rectangular slabs which are leaned against the pulley construction over the well shafts to create a tent- or roof-like construction protecting them against sand and dust being blown into the water of the well. An unusual example of a stepwell, whose entrance and well shaft are flanked by the same kind of minaret-like pillars, is to be found in the north of Gwalior where the Agra-Bombay Road meets the outer city ring road.

²¹ These have been discussed in Chapter Four.

seen adjacent to the Madana Mohana Temple at Bishnupur, in front of the Sūrya Temple at Konarak (Konārak, Konārka) in Orissa, and in the north-western corner of the Rāmacandra Temple compound at Raipur. Many temples, such as the Bṛhadiśvara Temple at Tanjore, have several wells used for different occasions or rites. The water from the Suna or Sona Kuāṃ (Golden Well) in the Jagannātha Temple at Puri, for example, is used only for the sacred bath of Lord Jagannātha on the festival of Snāna Tatra.²² Of the twenty-two so-called *tīrthas* in the Temple at Rameshvaram twenty are wells, and even the tank and the *kuṇḍa* are used as wells in that water is drawn up from them in buckets and poured over the heads of the pilgrims. In temple compounds, wells are frequently found very close to the sanctum in the innermost enclosure. The well nearest to the sanctum of the Kālahastīśvara Temple at Kalahasti is called the Pātāla Gaṅgā, meaning the 'Underground' or 'Subterranean' Ganges; this alludes to notions discussed earlier, particularly in Chapters Two and Three, whereby the Ganges is believed to emerge on special occasions at sacred places in other parts of the Indian subcontinent. The well in the inner enclosure of the Sundaresvara Shrine at Madurai is called the 'Goddess Well', and the water which priests raise from the well next to the sanctum of the Nāgarāja Temple at Nagercoil is sprinkled over the worshippers' offerings to purify them before they are presented to the god. Similarly, many Islamic mosque and tomb complexes, such as the Ṭānkā Masjid at Dholka and the Jāmi Masjid with Shaykh Salīm Chishtī's tomb at Fatehpur Sikri, have wells, which either tap the ground water or underground water cisterns, called *ṭānkā* or *barkā*.²³ An example of a rock-cut draw well connected with the Jaina faith can be seen at Cave No. 1 at the Jaina caves on the south-east side of the fort at Gwalior.²⁴

Sometimes, ordinary draw wells, where the well shaft is the only element penetrating the ground, were expanded to include intricate constructions above ground level. In such monuments, the well is usually contained within a highly raised platform reached either by steps or a section of raised ground positioned on one side of the well. The raised terrace surrounding the well is used for washing, bathing and massaging, and excess water is channelled off the terrace and collected in storage reservoirs. In order to provide quick and easy access to people who merely want to fetch water, a second lower access to the well has sometimes been provided via a room underneath the terrace contained within the platform. These rooms provide access to the water through a window in the side of the well shaft.²⁵ In more elaborate examples, a roof is frequently constructed over the platform containing the well to provide shade to people working on the terrace (Fig. 29 a). Wells belonging to this group are typical of the region surrounding Benares. Probably the most elaborate example of this kind is the well called the Yūpa Sarovar in the north of Benares, which has three storeys above ground (Fig. 29 b). The well has a high square terrace providing a lower and an upper access to the well, and a further storey has been built on top of the roof above

²² G. M. & B. Tripathy, 1992, *Sri Jagannath Puri: Past & Present*, p. 110.

²³ For a discussion of storage tanks under mosque courtyards see Chapter Four, 'Tanks'.

²⁴ This well is called the Ek-Patthar-kī Bāvlī meaning 'the one stone well'. Its name seems to refer to the fact that it is a monolithic rock-cut structure.

²⁵ P. D. Couté & J. M. Léger, 1989, *Bénarès—Un Voyage d'architecture—An Architectural Voyage*, pp. 90–91, 118–119.

the terrace.²⁶ The well edifice, which has open arcading on all sides, provides direct access to the well from below, space to work near the well in the middle, and a cool area for resting and socialising above the well. While the well shaft reaches down into the underworld, the three-storeyed edifice points up to the sky, and its four open sides face the four cardinal directions, thus establishing strong cosmological links.

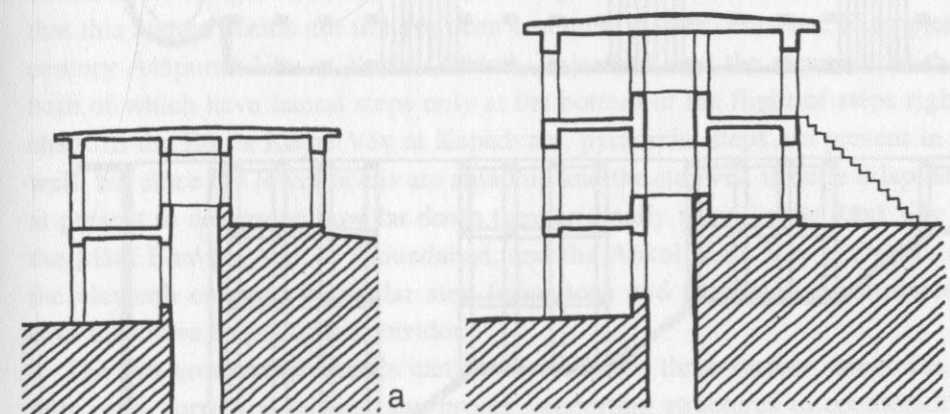


Fig. 29. Covered wells with two (a) and three storeys (b), providing access to the water on a lower and an upper level

Stepwells

In a stepwell, the vertical well shaft, also present in ordinary draw wells, is approached on one side by a long stepped corridor (*nāla*), which leads, frequently over several storeys, from the ground level down to the water of the well (Fig. 30). The majority of stepwells are oriented towards the cardinal directions. The well shaft can still be used to simply haul up water, but the steps also provide direct access to people who descend the steps to reach the water of the well. Frequently, there is a small square or octagonal tank in front of and linked to the well at the bottom of the steps which collects surplus water from the well. The basins are surrounded by a narrow terrace and are intended for washing and bathing, to keep the water of the well clean and potable (Plate 204). The area around and above the basins is open to the sky and ancient texts on architecture refer to them as *aṅgaṇa* (Sanskrit 'courtyard').²⁷ Around the shafts of these basins are platforms with parapet walls on each of the storeys which are frequently elaborately carved. *Aṅgaṇas* are usually found in larger stepwells only.

The main function of stepwells is to supply water, but because stepwells frequently have galleries built along the long corridor of steps and around the storage basins, integrating stone benches with sloping backrests into their design, they are also used as cool retreats from the summer heat by villagers and travellers. Particularly in desert areas and dry regions with few

²⁶ Ibid., p. 60; N. Gutschow & A. Michaels, 1993, *Benares: Tempel und religiöses Leben in der heiligen Stadt der Hindus*, p. 78.

²⁷ Jain-Neubauer, 1981, p. 3.

forests and shading trees, they were essential particularly to caravans. Stepwells are found at regular intervals along caravan routes where they provided water and cool resting places to travellers and their animals who, to avoid the heat of the day, frequently travelled by night.²⁸

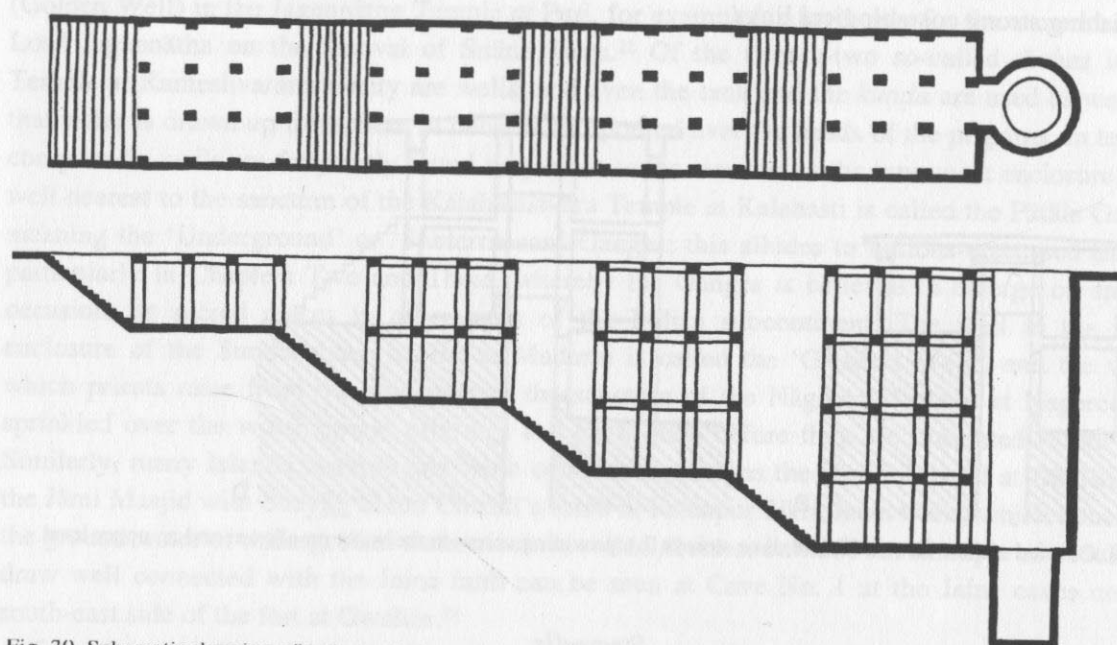


Fig. 30. Schematic drawing of a stepwell

While there are rock-cut stepwells which date from the second to fourth centuries AD,²⁹ the earliest structural stepwells were probably constructed in the early seventh century AD.³⁰ Most of them, however, date from a period between the twelfth and the seventeenth century. The tradition of stepwell architecture is indigenous to India, but was adopted and refined by the Muslim rulers who constructed stepwells in north-western India, particularly during the fourteenth and fifteenth centuries. While their basic design was not altered, in many cases the use of stepwells changed from being a Hindu religious and public monument, to becoming predominantly an Islamic private palace structure.³¹ There are, however, exceptions to this rule, such as the Islamic Jhālra Stepwell at Fatehpur Sikri, which is located in a public place and cannot be classified as a palace structure. Stepwells are found in many parts of India, and are typical of regions with long, ex-

²⁸ Gutschow & Pieper, 1978, p. 317.

²⁹ One example of such an early stepwell is the Nāvghān Kūo at Uparkoṭ in Junagadh whilst others, dating from the same period, may be seen in the area around Porbander (Jain-Neubauer, 1981, p. 19).

³⁰ The three earliest known structural stepwells are the Bochavdi Vāv near the village of Dhank (Dhānk) in Gujarat, and the Jhilanī (600 AD) and the Mañjuśrī Vāvs (early seventh century AD) in Dhank itself.

³¹ Jain-Neubauer, 1981, p. xv.

tremely hot, dry summers. Therefore they are not typical of Nepal or Sri Lanka, where the climate is more balanced and water is available all year round.³²

The steps in stepwells are usually straight and parallel, but some also integrate pyramidal formations into their flights of steps, a form which is more closely associated with *kunḍas*. Through the addition of such lateral steps, the *nāla* can be steeper and therefore shorter. It is noteworthy that stepwells with triangular step formations, at least if also present in the upper part of the *nāla*, are generally quite early, dating from about the eleventh and twelfth centuries, and that this feature seems not to have been common in later centuries. Examples are the late twelfth-century Āśāpurnā Vāv at Nadol (Nādol), Rajasthan, and the stepwells at the cenotaphs at Ahar, both of which have lateral steps only at the bottom of the flight of steps right in front of the well shaft. In the Bātris Koṭha Vāv at Kapadvanj, pyramidal steps are present in the upper part of the well, but since the lower levels are silted up and the stepwell is quite dilapidated, it is not possible at present to determine how far down they originally went. In the Rāñī Vāv at Patan (Plate 207), the Mātā Bhavānī Vāv at Ahmedabad, and the Añkol Mātā Vāv at Davad, all three dating from the eleventh century, triangular step formations and parallel steps alternate at regular intervals over the entire length of the corridor.³³

The sub-group of stepwells can be divided into three further categories. There are stepwells with open corridors, lacking any proper supporting structures to counteract the inward thrust of the side walls, stepwells with L-shaped corridors, and some which are more enclosed by having pavilions and post and beam constructions to strengthen the long vertical side walls of the stepped corridors.

The largest and most widespread group of stepwells are those which have an open stepped corridor leading down to the water in the well, and do not need internal constructions to strengthen the side walls. In many cases, this has been possible because the ground is hard and rocky, providing sufficient strength to the vertical side walls of the corridors. In addition, these stepwells are frequently comparatively shallow and therefore require only a short flight of steps. Lastly, the side walls of such stepwells are frequently horizontally stepped and widen slightly towards the top. These represent stable constructions which do not require additional internal support structures. Such open stepwells are distinguished from linear *kunḍas* by having stepped corridors which are clearly separated from the well shafts by a screening wall or archway at the end. The wall which separates the well from the *nāla* seems intended more to mark the end of the steps and the beginning of the well than to strengthen the construction. There are a few very early examples of this sub-group of stepwells, such as the Nāvghān Kūo and the Aḍī Kaḍī Vāv (Plate 205), both rock-cut structures in the fort of Junagadh, whose original parts date from about the

³² Stepwells are, however, not exclusively found in dry or desert areas, as the stepwell in the Kaṭ Golā complex at Murshidabad shows.

³³ An unusual example of this group is the circa fifteenth century stepwell at Neemrana (Nīmranā) in Rajasthan, but here only one triangular step formation is found at the very bottom of the flight of steps.

second to the fourth century AD; they were expanded in the eleventh century.³⁴ A further example is the stepwell opposite the Archaeological Museum at Lakkundi which seems to predate the earliest eleventh-century temples at the site.

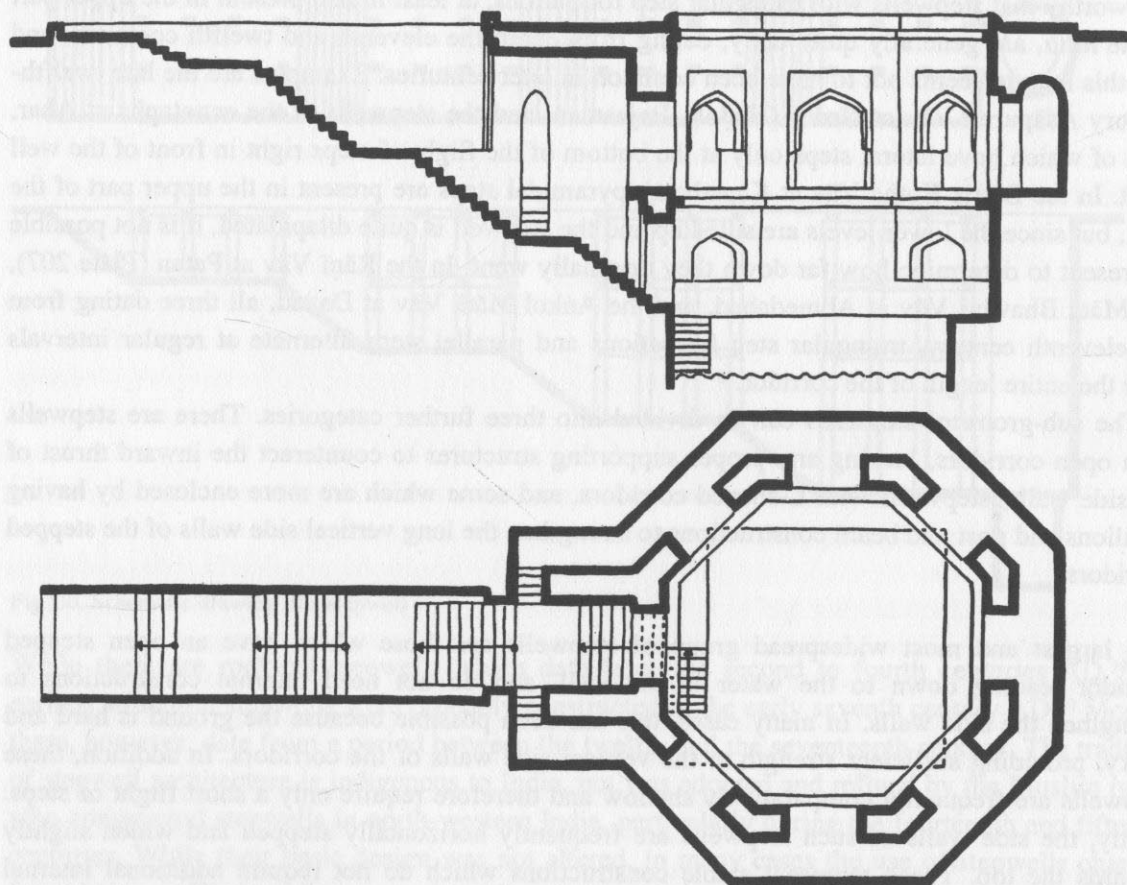


Figure 31. Malpannagudi: simple stepwell with short flight of steps

The simplest examples of this open stepwell sub-group have very short flights of steps of only about two metres in length leading to small square wells, such as the one behind the Ibrahim Rauzā (1626 AD) at Bijapur, and the well north-east of the Kuśavart Tīrtha at Trimbak. Larger versions, although still simple in layout, are the stepwells at Malpannagudi (Malpannagudi) near Vijayanagara in Karnataka (Fig. 31), and the Jhālārā directly adjacent to the Buland Darvāzā of the Jāmi Masjid (1572 AD) at Fatehpur Sikri (Plate 206). The well at Malpannagudi has been dated to

³⁴ Jain-Neubauer, 1981, fig. 26; G. Michell, 1990(a), *The Penguin Guide to the Monuments of India: Buddhist, Jain, Hindu*, vol. 1, p. 292.

1412 AD,³⁵ and the Jhālārā is contemporary with the adjacent mosque.³⁶ The latter well used to be surrounded by a colonnade above ground, of which little survives, while the well at Malpannagudi has an arcade just below the ground, relating it also to spiral-wells, discussed later in the chapter. The octagonal well shaft of the Jhālārā is quite large in comparison with its narrow stepped corridor, and is therefore also related to the tank-stepwell hybrids discussed in Chapter Four.

The stepwell character becomes more pronounced, the smaller the diameter of the well shaft in relation to the width of the corridor and the longer and deeper the flight of steps penetrating into the ground. A clearer example of a stepwell is the one in front of the platform bearing the Kaṇḍarīya Mahādeva and the Jagadambī Temples at Khajuraho (Plate 208).³⁷ It has a beautifully carved lintel above the end of the stepped corridor showing *haṃsas*, men, horse-riders, elephants and lotus ornamentation. The well shaft is surrounded by a wide platform, more common with draw wells. Even clearer examples, where the well shafts are not wider than the stepped corridors, are the well at the cenotaphs at Ahar outside Udaipur and the Jetānī Vāv in the fields near Ghumli. A larger and wider stepwell with stepped side walls in five tiers is the Gandhak-kī Bāvlī at Delhi (Plate 209), whose name derives from the smell of sulphur (*gandhak*) in its waters. The stepped corridor is about forty metres long and twelve metres wide. In front of the well, which is hidden behind a screen with a pillared gallery, is an approximately twenty metres long water basin. The stepwell is believed to have been built in the reign of Iltutmish (1211–36 AD).³⁸ A similar example also from Delhi is the Ugrasen-kī Bāvlī which dates from the late Tughluq or Lodī period.³⁹ It is about sixty metres long and fifteen metres wide, and was constructed in four tiers (Plate 210). Between the well in the north and the basin in front is a screen with two pointed arches one above the other, the one at the bottom open, and the one above more in the shape of an *īwān* derived from mosque architecture. In this example, the long side walls of the corridor were reinforced with blind arches. While the second level from the top and the lowest level only have decorative recesses, there are cool apartments behind the open pointed arches on the third level. Integrated into the south-western corner of the stepped corridor is a small mosque. The seventeenth-century stepwell of Bīr Singh Dev⁴⁰ in the village of Sirol, eight kilometres from Datia, is built in two large tiers, and has arcaded galleries along the sides of the stepped corridors, as a partition in front of the well and surrounding the well shaft. There are small recesses in the side walls which used to hold lights. More complicated examples with apartments on various levels, both at the sides and at the end of the steps are the Rājōn-kī-Bāin⁴¹ at Delhi, the stepwell in

³⁵ J. M. Fritz & G. Michell, 1991, *City of Victory: Vijayanagara, the Medieval Hindu Capital of Southern India*, p. 111.

³⁶ A. Petruccioli, 1988, *La Città del Sole e della Acque: Fatehpur Sikri*, p. 56; S. A. A. Rizvi, 1992, *Fatehpur Sikri*, p. 72.

³⁷ The Kaṇḍarīya Mahādeva Temple is also frequently spelled Kandāriyā Mahādeva Temple.

³⁸ Y. D. Sharma, 1964, *Delhi and its Neighbourhood*, p. 61.

³⁹ *Ibid.*, p. 100.

⁴⁰ In colloquial language, he is usually referred to as Bir Singh Deo.

⁴¹ This stepwell is at times also spelled Rājōn-kī-Bāin. It seems to derive its name from having been used by masons (*rāj*). In the north-western corner of the arcade of the well is a mosque and a *chattrī*. The latter bears an inscription dating the complex to 1506 AD. Small mosques are also closely associated with other stepwells, such as the Ugrasen-kī Bāvlī at Delhi and the Dādā Harīr Vāv at Asarva (Āsarvā) in Gujarat.

the Barā Imāmbārā at Luknow (214), and the Purāṇa-Ghāt-Kanyā Stepwell (Plate 212), also called Ghāt-kī-Gūnī Bāvlī, at Jaipur. In the so called Bāvlī, an open stepwell in Naugaon (Naugāmv) outside Dhar, apartments were only integrated into the wall section separating the open corridor from the well-shaft behind. It is, however, notable that this section resembles a risalit, a feature discussed in detail in connection with *kunḍas* in the previous chapter. Similar to many *kunḍas* with risalits, the well shaft in this stepwell was integrated into the protruding wall element and the rooms located in between the open *nāla* and the covered well-shaft remain cool and comfortable, even in the summer heat.

There are a few unusual examples of stepwells with open corridors, which have been constructed on a cross-shaped ground plan and have a relatively large central square well and narrow open flights of steps in the centre of all four sides. An example from Gujarat is the Caumukhī Vāv (Four-faced Well) at Chobari in Gujarat. Its plan corresponds to the fourth class of the *sāstric* description of stepwells, called *vijaya*.⁴² The walls of the well and those of the stepped corridors have small recessed niches containing images of the *saptamātṛkās* and other divinities. An example of a similar layout, where the long stepped corridors, however, turn sideways in the middle to resemble the shape of a swastika, is the Swastika Well located behind the temple at Tiruvellarai (Plates 28–29). There are three stone beams, set at regular intervals one above the other, at each of the four openings into the side of the well-basin created by the stepped corridors. The top two beams show elaborate figural carvings of the *saptamātṛkās*, Śiva and Pārvatī, Nara-simha and others. In both these examples, the flights of steps spread out into the cardinal directions from the central well, and seem to visualise the spread of creation from a central point. That the religious connotations of these wells take precedence over practical considerations is evident from the narrow and bent corridors which do not lend themselves easily to the fetching of water, from the fact that neither well has a pulley mechanism, and from the elaborate religious carvings found at both sites.

The second group of stepwells are constructed on an L-shaped plan, where the stepped corridor does not run in a straight line, but turns at a right angle. While in some stepwells, the two parts of the corridor are of equal length, there are others, where the part nearer to the ground level is shorter. Although J. Jain-Neubauer has suggested that this type “seems to be exclusive to Gujarat, as it is found nowhere else”,⁴³ the following discussion of monuments will show that this sub-group is in fact spread quite widely over the Indian subcontinent. There are a series of early examples for this sub-group, such as the rock-cut Mañjuśrī Stepwell at Dhank (Dhāṅk) in Gujarat, and the rock-cut well at Mandor near Jodhpur, both dating from the early seventh century AD. A stepwell at Choti Khatu in the Nagaur district of Rajasthan, dates from the eighth century AD.⁴⁴

A simple L-shaped stepwell on a very small scale, used for ablutions, is located in the temple complex at Siddhpur north of the Bindu Sarovar tank. The short flights of steps are extremely narrow and have high walls. There are no supporting constructions against the thrust of the side

⁴² Jain-Neubauer, 1981, p. 68.

⁴³ Ibid., p. 27.

⁴⁴ Mankodi, 1991, p. 235.

walls. A similar but slightly larger well from Maharashtra belongs to the eighteenth century Gr̥ṣneśvara Temple at Ellora. A short flight of steps leads to a first landing where it meets a corridor of double the length set at ninety degrees. There is also an ordinary draw well to the north of the temple. A further example from Maharashtra, where the difference in length between the two flights of steps is less extreme, is the brick-built Hāthī Bāvlī (Elephant Well), which allegedly was used to water elephants, outside Adavad. The well is adjacent to the Lāl Bāgh enclosure and probably also supplied the water for the irrigation of the garden. An example from Karnataka is the tenth-century AD Vāpī No. 1 at Sirwal (Sirvaḷ), which has two short flights of steps of equal length.⁴⁵ L-shaped stepwells are frequently quite small and simple without additional basins for bathing, and usually without pavilions or complicated constructions against the inward thrusts of the side walls. They are often found in villages and on country roads. Slightly more elaborate examples of L-shaped village stepwells are the Hari-Ūṃ-Āśram Bāvlī at Desuri (Desūrī) in Rajasthan (Plate 211) and the Nadir-kī Vāv in Nagaka Gau (Nāgakā Gāmv) near Ghumli. In the latter example, the corridors are surrounded by a terrace about one and a half metres high, a common feature of draw wells but unusual for stepwells. The entrance to the well, which is approached by a flight of steps, is flanked by lateral gate-towers and, at the end of the first *nāla*, where the corridor turns to the north, two shrines flank the corridor at ground level. To the north of the well shaft is a large cattle trough. A more complicated example on a larger scale is the Jñān Vāv to the west of the main Mūl Dvārakā Temple complex at Visavada. It has a short flight of steps running from south to north, and a longer corridor leading from the west down to the well shaft in the east. The Jñān Stepwell has an entrance pavilion and, unusually for the L-shaped sub-group, three pavilions topped by pyramidal roofs were built into the stepped corridor to strengthen the construction internally (Plate 214). In the side walls of the stepped corridor are several niches containing images of Viṣṇu and Sūrya. To the north of the well shaft is a large trough to water animals.

Probably the largest and most ambitious projects of stepwells built on L-shaped plans are the so called Shāh Qulī's Bāvlī⁴⁶ at Fatehpur Sikri, located beneath the palace buildings on the main road to Agra and the Navalakhi Vāv in the Lakṣmī Vilās Palace at Baroda. Shāh Qulī's Bāvlī is the largest well at the site and probably the most elaborate Mughal example of a stepwell.⁴⁷ Its entrance is in the north, and a short flight of steps leads down to a landing, where it meets the long main stepped corridor, lying on an east–west axis with the draw well at the eastern end (Fig. 32). The two corridors are open and the well descends over four storeys. The upper storey has pillared galleries along both sides of the long corridor, similar to examples at Delhi discussed in the last paragraph. The level underneath it accommodates two large water wheels (*rehaṇṭs*) by which water was raised out of the well and stored in a separate deep basin to the north-west of the

⁴⁵ Ibid., p. 235; M. W. Meister & M. A. Dhaky, 1986, *Encyclopaedia of Indian Temple Architecture: South India, Upper Drāviḍadēśa (Early Phase, A.D. 550–1075)*, p. 156–157.

⁴⁶ The well is believed to have been built by Shāh Qulī, whose ruined palace is located on the opposite side of the road (Rizvi, 1992, p. 88; P. Davies, 1989(a), *The Penguin Guide to the Monuments of India: Islamic, Rajput, European*, vol. 1, p. 218).

⁴⁷ R. Nath, 1971, “Account of a typical Mughal Step-well (bāvlī) and a Well-house (kūpāgār)”, p. 133.

stepwell. There is a room which is open towards the storage basin, and two octagonal chambers are located on the north and the east sides of the well shaft providing cool apartments in the heat of summer. Thus the well supplied water to the palaces but could also be used as a cool retreat. Unusually for stepwells, the pillared corridors run also around the octagonal well shaft on the up-

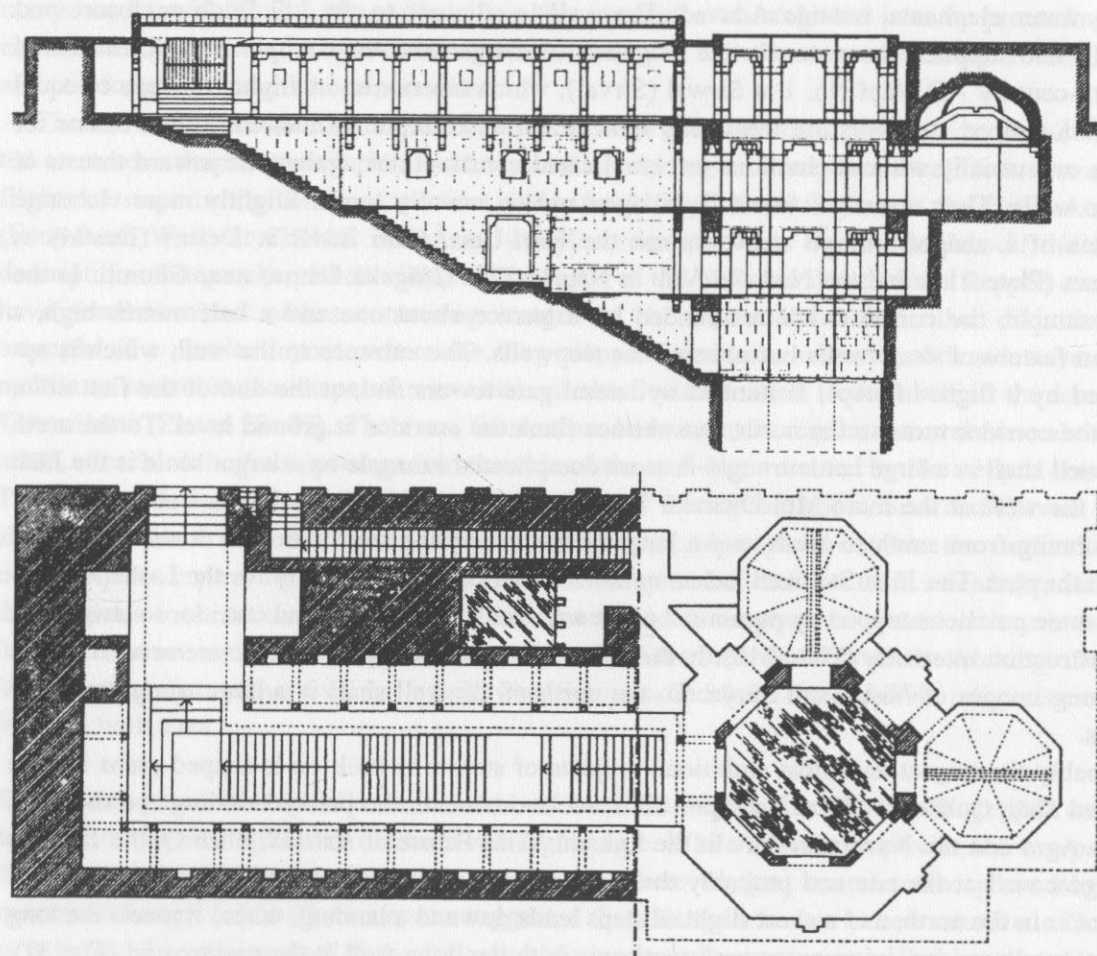


Fig. 32. Fatehpur Sikri: the L-shaped Shāh Qulī's Bāvlī (Petrucchioli, 1988)

per two storeys, a feature typical of spiral-wells.⁴⁸ Further exceptions, having straight corridors and galleries surrounding the well shafts, are the Ra Khengar Vāv, also called Khingar Vāv, at Vanthali, the Indaravālī Bāvlī at Fatehpur Sikri,⁴⁹ Bīr Singh Dev's Bāvlī at Sirol and the stepwell of the Barā Imāmbārā at Lucknow (Plate 213). These have a straight corridor and a gallery surrounding the well shaft. The L-shaped fifteenth-century Navalakhi Vāv at Baroda has four pavilion towers and four intermediate supporting frameworks against the thrust of the side walls; it

⁴⁸ Spiral-wells will be discussed later in this chapter.

⁴⁹ For plans and sections of these two wells, see J. Jain-Neubauer, 1981, plan 6, and A. Petruccioli, 1988, p. 99.

penetrates five storeys deep into the earth. The angle where the nine metres long stairs of access meet the thirty-five metres long main corridor is topped by a domed pavilion. Undoubtedly, this stepwell is one of the largest and most elaborate examples of the L-shaped sub-group. Although today the dilapidated Bīja Maṇḍal Bāvlī at Vidisha looks L-shaped and K. Mankodi identifies it as such,⁵⁰ it does not seem to have been planned as such. In this example, a further short flight of steps (today in ruins) located opposite the stairs of access in the south, seems to have led to the main corridor of the stepwell.⁵¹ This defines the construction as a stepwell with a straight corridor and two access stairs leading down to the main *nāla*. In another variation, the stepwell south-east of the Rānī Mahal at Chandragiri (Candragiri) in Andhra Pradesh has an additional bend in its corridor. The first flight of steps leads from the east to the west, until it meets the second corridor running from the south to the north at an angle of ninety degrees. From the landing in the north, the steps turn again at a right angle to run parallel to the first corridor and lead to the well shaft in the east.

K. Mankodi argues that L-shaped wells represent an earlier stage of stepwell architecture, in which the builders had to divide the corridor into two sections because they were as yet unable to construct an extended and straight flight of steps.⁵² However, the length of the corridor does not seem to influence the stability of its side walls, which is more dependent on the local soil condition and the presence of supporting beams or pavilions. The earliest L-shaped stepwells are rock-cut structures which, from a constructional point of view, could equally have been straight; and the main corridors of Shāh Qulī's Stepwell at Fatehpur Sikri and of the Navalakhi Vāv at Baroda are much longer than many straight stepwells. The latter two examples, dating from the sixteenth and the fifteenth centuries also show that the construction of L-shaped stepwells is not restricted to early periods but continued alongside the construction of stepwells with one straight *nāla*. The choice of an L-shaped plan seems to have had more to do with issues concerning the available plot of land. Stepwells which go very deep and therefore have long corridors need large pieces of ground. The long wedge-shaped *nālas* can be shortened by splitting the corridor into two and turning the second flight of steps to one side, a feature already encountered in linear *kunḍas*. The L-shaped corridor of the Mughal stepwell at Fatehpur Sikri, for example, neatly fits around the storage tank adjacent to the well. A further reason for creating a bend in the stepped corridor may have been to prevent dirt from being blown into the well via the steps. In addition to these practical considerations, religious aspects should also be taken into account. The well shaft is the most sacred part of the well monument. Through a bend in the steps the shaft is not directly visible from the top of the steps, and is therefore approached more respectfully. Because the entrance to the well is also not visible from the bottom of the steps, the person descending into the well really embarks on a journey into the nether regions, where the world above is merely reduced to a small section of sky seen through the well shaft above.

⁵⁰ Mankodi, 1991, p. 235.

⁵¹ This corridor leads from the west to the well shaft in the east.

⁵² Mankodi, 1991, p. 236.

Because the ground in most parts of Gujarat is loose and sandy and easily slips away, it is typical of Gujarati stepwells to have structures supporting the side walls of the corridor. These can be pavilions (*kūṭa*)⁵³ built on top of each other into the narrow corridors (Plate 216), a simpler framework of lintels and beams (Plate 217), or a combination of both methods (Plate 218). Through these additional constructions stabilising the walls on either side of the flight of steps, the corridors could be made wider and penetrate deeper into the ground. Additionally, the supporting constructions create protected and shaded areas for rest, and further emphasise the multi-storeyed nature of the well monuments, which are complicated edifices under ground. Stepwells with supporting constructions need not necessarily be larger and more complicated constructions, as is shown by the simple Derānī Vāv near Ghumli, the stepwell at Uvarsad (Uvārsad) (Plate 217) in Gujarat, and the Bīja Maṇḍal Bāvlī (circa eleventh–twelfth century) at Vidisha. All three simply have beams, some of them supported by pilasters, which firmly hold the side walls apart. It is interesting to see that although the cross-beams in the Bīja Maṇḍal Bāvlī are broken and only small stumps remain, the side walls still remain intact (Plate 236). The Bīja Maṇḍal Bāvlī is located to the north-east of the former circa eleventh to twelfth-century Paramāra Temple, which in the seventeenth century was converted into a mosque by Aurangzeb. The stepwell at Uvarsad has stone beams, while the rest of the structure is built of brick. The brick walls were plastered and decorations, similar to those carved on the walls of stone built examples, were painted in dark red-violet vegetal colours on a white background.⁵⁴

Examples of narrow stepwells containing intermediate tower-like pavilions as strengthening features, are the tenth to eleventh-century stepwell at Modhera (Plate 219) and the Vajināth Vāv at Kapadvanj. The early twelfth-century Vikia Vāv at Ghumli, the Vidyādhara Vāv (AD 1487) at Baroda and the Gaṅgā Vāv at Wadhwan (Vadhvān) all in Gujarat (circa thirteenth to fourteenth century) still have narrow corridors, but the beams of the pavilions do not simply rest on wall pilasters but are further supported by a pair of pillars set slightly away from the vertical walls and into the passage. Because the pillars are still very close to the sides of the corridor, they do not actually create three openings to walk through but merely convey the illusion of a wider corridor. The three pavilion towers in the Vikia Vāv have been set unusually wide apart from one another, a sign that the ground must be quite hard and stable. This arrangement admits a substantial amount of light and air into the stepped corridor.

Most typical of this sub-group, however, is a wider corridor with pilasters at the sides and two columns set further away from the walls to create three trabeate openings supporting three beams above. Because the positioning of the columns is related to an octagon, a standard type of planning in Gujarat, the central opening is slightly wider than those on the sides (Plates 220). Probably the best known examples of this group are the eleventh-century Mātā Bhavānī Vāv at Ahmedabad, the fifteenth-century Dādā Harīr or Harīr's Vāv (1499 AD) at Asarva (Āsarvā), also

⁵³ The term *kūṭa* is at times also spelled *kūtā*.

⁵⁴ Wall paintings of this kind are typical of the region and can for example also be seen on the walls in the house where Mahātmā Gāndhī was born in Porbander.

known as the Bāi Harīra Vāv,⁵⁵ and the sixteenth-century Ruda Vāv at Adalaj (Adālaḥ), all three in Gujarat. The last three examples all have separate basins for washing and bathing in front of their well shafts. The galleries above the opening are profusely carved and the parapet walls consist of stone benches with sloping back rests. The Dādā Harīr Vāv, and the Ruda Vāv descend for five storeys under ground, and both were commissioned by female patrons. A lesser known example of a wide stepwell with extra columns is the Navalakhi Vāv (1405 AD) in the garden of the Lakṣmī Vilās Palace at Baroda, which makes use of both kinds of supporting structures, intermediate framework made only of beams, and sections with towers of piled up pavilions. In the Rānī-jī-kī Vāv⁵⁶ at Bundi the columns are tall and slender, with flying cusped arches, creating rather a delicate screen than a strong load-bearing construction. An interesting feature of the circa mid-nineteenth-century Jethābhāi Vāv at Isanpur (Isānpur) near Ahmedabad is that the central openings of the two top pavilions nearest to the well have, in addition to the beam bridging the opening above, a true voussoir arch set between the two pillars, while stepwell constructions are usually purely trabeate. The Ra Khengar Vāv (circa 1230–1235 AD) at Vanthali has a large slightly pointed arch bridging the entire width of a wide corridor with two wall pilasters and two pillars. Whilst narrow single span corridors are frequently stabilised by arches, this is an unusual example of such a wide arch.⁵⁷

Probably the widest and largest stepwell of all, which has wall pilasters and six intermediate columns, is the Rānī Vāv, the Queen's Stepwell, at Patan (Plate 221). It was begun by Queen Udayamati in 1063 AD and never completely finished.⁵⁸ It has six storeys, measures sixty-five metres in length, and contains over five hundred sculptures along its corridor walls, brackets and beams. For the last few centuries, this probably most ambitious stepwell construction remained covered by large deposits of sand from the nearby Sarasvati River; it was excavated and desilted only in 1986. An even deeper stepwell with seven storeys, which however has mainly plain walls and simple pointed arches, is the fifteenth-century stepwell in the village underneath the fort of Neemrana (Nīmraṇā) in Rajasthan.

In stepwells which have pavilion towers constructed over the steps in the narrow corridors, the roofs of the top pavilions are usually level with the surrounding ground (Plate 222). In only a few cases, a further layer of pavilions, protruding over the surrounding ground and accessible from the sides like a bridge, was added on top. In some stepwells, only one such pavilion was constructed above the steps, right at the end of the corridor in front of the well shaft. Examples are the stepwell in the Kaṭ Golā complex, the palace area of a wealthy Jaina merchant at Murshidabad, and the Trīkam-Bārot-nī Vāv at Patan in Gujarat. In other cases, such as the Mātā Bhavānī Stepwell at Ahmedabad, the Vikia Vāv at Ghumli, the Jñān Vāv at Visavada (Plate 223), and the

⁵⁵ The second-named stepwell was constructed by Bāi Harīr Sultānī, locally known as Dādā Harī or Harīr. Today, Asarva is on the outskirts of Ahmedabad.

⁵⁶ This well is also commonly known as the Jānī-jī-kī Bāvlī.

⁵⁷ A rare example of a stepwell with four columns and two wall pilasters on the highest level, and of two columns and two wall pilasters on the level below it, is the Dūmgriyā-kī Bāvlī at Ghanerao.

⁵⁸ Mankodi, 1991, p. 30.

Bātris Koṭha Vāv at Kapadvanj,⁵⁹ a whole series of such raised pavilions reach out above the ground. Although such wells consist overwhelmingly of a negative or inverted form of architecture under ground, they are still visible above ground. Yet at surface level they reveal only very little of their subterranean splendour. In this way, stepwells located on trading routes in the countryside were more easily seen from a distance.⁶⁰ Both the stepwell at Modhera and Vāpī No. 2 at Sirwal have a small shrine built over the stepped corridor which protrudes above the surrounding ground (Plate 219). The effect of marking a subterranean edifice above ground was also achieved by adding small *chatrīs* on the sides of well monuments, such as those at the Dādā Harīr Stepwell at Asarva (Plate 224) and the Rānī-jī-kī Vāv at Bundi, and by constructing elaborate entrance pavilions, which will be discussed in detail later in the chapter.

There are a few interesting examples where the stepped corridors of stepwells are entirely roofed over, and where only the first few steps and the well shaft remain open and visible as holes in the ground. Such wells can be used, like other stepwells, to draw up water and to reach the water via the stairs. It is, however, less common in this sub-group of stepwells to have underground rooms or galleries functioning as cool retreats. An example of a covered stepwell is the late fifteenth or early sixteenth-century Bāvlī in the Dīvān-i-Ām quadrangle adjacent to the Salīmgarh in the fort at Agra (Plate 225).⁶¹ The underground stepped corridor is not entirely straight and turns slightly east after about ten metres to meet the large brick built well shaft. A second example is the stepwell of the Bāvlī Sāhib Gurudvārā at Goindwal (Plate 226). The well, which is situated in the courtyard of the gurdwara, was completed in 1559 AD and seems to predate the construction of the complex.⁶² It was constructed as a water place along the route from Delhi to Lahore, and it is believed that Guru Amar Dās received orders for its construction from Guru Nānak while in a trance. The well has prominent pavilions over the entrance to the stepped corridor and covering the well shaft. The covered corridor measures about fifty metres in length, and it is interesting that although the corridor is entirely hidden under ground, its position is clearly marked by white lines painted on the courtyard. A further example of this completely covered corridor type is the Siṃhakiṇaru Stepwell in the compound of the Bṛhadiśvara Temple at Gangaikondacholapuram (Plate 227). By being covered, these wells can be securely locked and protected, as is the case in the Tamil example, which is used only for the washing of the main temple image. At Goindwal, the devotees resort to a complicated rite of climbing the steps of the well while uttering prayers. They always climb only one step higher than before, and then descend again to the very bottom of the steps, before ascending to the step above the highest reached before; so they continue until they reach the top. Here, the corridor has been covered to enhance the feeling of having entered a different world in which penance is done for one's misdeeds and

⁵⁹ Today, the Bātris Koṭha Vāv at Kapadvanj is so closely surrounded by modern houses that access to the top pavilion is obstructed.

⁶⁰ The same idea was encountered earlier in this chapter in connection with raised draw wells which are surrounded by tall towers, a phenomenon well known from the Shekavti region and other areas of the subcontinent.

⁶¹ R. Nath, 1985, *History of Mughal Architecture*, vol. II, p. 141.

⁶² P. S. Arschi, 1986, *Sikh Architecture in Punjab*, pp. 74–75.

where an arduous climb is necessary to leave the dark and hot corridor behind and reach daylight and fresh air again.

Of all types of water structures, stepwells are the group that has attracted most attention from art-historians. Until now, however, all studies of stepwell architecture have concentrated on the north of India in general, and on Gujarat in particular, regarded as the homeland and sometimes even as the only place where stepwells exist. R. Nath examined stepwells at Agra and Fatehpur Sikri, but believed stepwell architecture to have been "introduced in the Mughal architecture by the artisans of Gujarat",⁶³ maintaining that they were less important imitations of Gujarati stepwells.⁶⁴ At Fatehpur Sikri, where many buildings exhibit a whole range of architectural influences from Gujarat, it may be legitimate to argue in favour of direct Gujarati influence. But for most other regions, such as West Bengal, Maharashtra, Madhya Pradesh, Karnataka and Andhra Pradesh, to mention just a few other areas with clear examples of stepwell architecture, this would seem unjustifiable. J. Jain-Neubauer has argued that L-shaped stepwells are unique to Gujarat, in spite of the fact that many examples exist throughout the Indian subcontinent. Such an opinion seems based on a preoccupation with Gujarati examples without the benefit of evidence to the contrary available from other regions. A closer examination of stepwells throughout India shows that only those with an elaborate intermediate framework of beams and pavilions are predominantly found in north-western India and that they are less typical of other regions. This has to do with the soil condition in that region, which needs more supportive constructions. The much larger group of stepwells with a straight corridor and no internal supporting framework against the sloping side walls, however, is practically non-existent in the north-western region, while it is found in almost all other regions of the subcontinent. Thus although stepwells are common in Gujarat, only a very small group of these well monuments—albeit one which undoubtedly shows a highly impressive development—is typical of this region. In this context, it may be more convincing to explain the presence of stepwells in other parts of India from a climatic point of view, rather than as the diffusion of a Gujarati phenomenon. Stepwells are generally typical of regions with extremely hot, dry, and long lasting summers, where ground water is often the only available source of water during the hot season, and where it was essential that shade was provided for travellers in a barren landscape.

Spiral-wells

In a spiral-well, the galleried apartments and the steps of access are arranged to surround the well shaft and spiral around it. In comparison with stepwells, which show a linear or longitudinal layout, spiral-wells are planned vertically. The *Viśvakarma Vāstu-śāstra* describes the staircase winding around the well shaft as being in the form of the coils of a snake.⁶⁵ A tendency towards this kind of well monument can be seen in the Shāh Qulī's Bāvlī at Fatehpur Sikri, the Ra Khengar Stepwell at Vanthali, Bīr Singh Dev's Bāvlī at Sirol, and the stepwell of the Baṛā Imām-

⁶³ Nath, 1971, p. 133.

⁶⁴ Ibid., pp. 132, 134.

⁶⁵ Jain-Neubauer, 1981, pp. 16–17; K. V. Sastri, 1958, *Viśvakarma Vāstuśāstram*, chapter 33, pp. 438–443.

bārā at Lucknow (Plate 215). These edifices are essentially stepwells but have prominent galleries surrounding the well shafts. Examples showing a more distinctive spiral-well shape, are the Helical Stepwell at Pavagarh (Pāvagarh) in Gujarat, the well in the Digambara Jaina Complex at Papora (Paporā) in Madhya Pradesh (Plate 228), and the Āssi Khāmbhā (Eighty-Pillared) Well in the fort of Gwalior. In all three examples, there are still straight stepped corridors, but they are very narrow and short, and continue spiralling around the inner side of the large wells. The upper level of the Āssi Khāmbhā, just below ground level, is an open gallery, and there is a further hall overlooking the well on the north side above ground (Plate 231). A similar hall was found on the side of the Baṛī Bāvlī at Golconda, and provided the kings and queens with a cool, shaded leisure-time environment.⁶⁶ Given that the area of the Āssi Khāmbhā is walled, protected, and situated adjacent to the Mān Mandir palace, it was probably used in a similar way. In the Champā Bāvlī,⁶⁷ on the northern side of the Muñj Tālāb at Mandu, apartments are found surrounding the base of the well (Plate 229). These are connected with a series of vaulted rooms known as *takhāna*, which are almost on a level with the lake feeding the well. The fourteenth-century so-called Circular Bāvlī in Firūz Shāh Kotlā, the citadel of Firuzabad (Firūzābād), the fifth city of Delhi, has apartments on two levels running around the large central well shaft (Plate 230). The upper level is above ground and the lower was built below ground level into the side of the well. A spiral-well at Mandwa (Māndvā) in Gujarat, has apartments on three levels opening onto a well shaft. Access to the upper chamber is provided via a passage spiralling around the well shaft (Fig. 33).

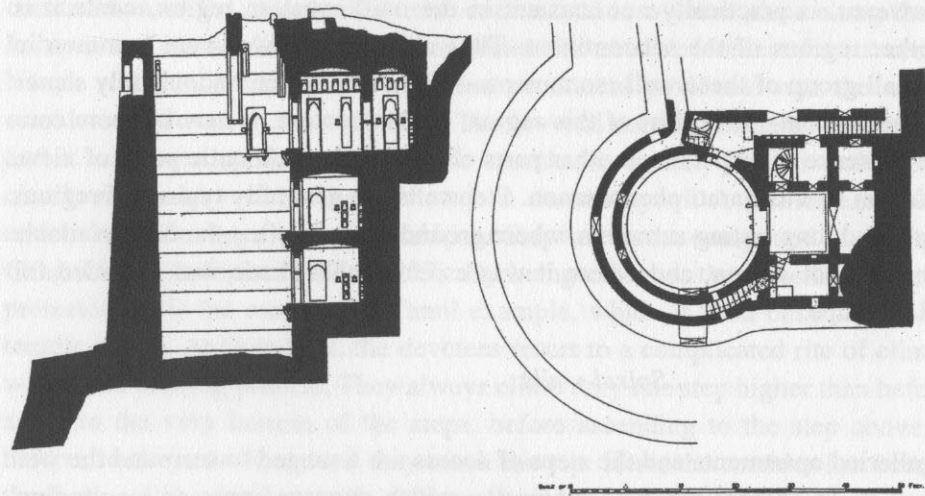


Fig. 33. Mandwa: section and plan of the spiral-well (Burgess & Cousens, 1903)

A more complicated arrangement is found in the fifteenth or early sixteenth-century Bhamaria (or Bhrahmarīā), a spiral-well outside the village of Sojali (Sojalī) near Mehmudabad (Mehmudābād)

⁶⁶ Commercial Book Centre, no date, *Golconda Fort & Qutab Shahi Tombs*, p. 10.

⁶⁷ Its name derives from the *champak* flower because the water is said to be as sweet as the flower's smell (D. R. Patil, 1992, *Mandu*, p. 32).

in Gujarat. The well was surrounded by four circular chambers above ground level which, however, have almost entirely gone. In the first storey under ground there are eight square rooms which open onto the octagonal well shaft by means of large *jharokhā* windows. Between the windows are two large parallel arches overspanning the well shaft which used to support the pulley mechanism. The steps spiral around the well shaft and further down, where rooms on a third level were probably used as a cool retreat in the summer heat.⁶⁸ Similar examples of spiral-wells where the apartments above ground level are still intact are the Andherī Bāvlī at Mandu and the Saṅgīn Bāvlī (Stone-masonry Well) near the Hāthī Pol (Elephant Gate) at Fatehpur Sikri (Plate 27). The well is surrounded by four chambers above ground level and by four below, and has four staircases which spiral around the well without crossing each other (Fig. 34). Two

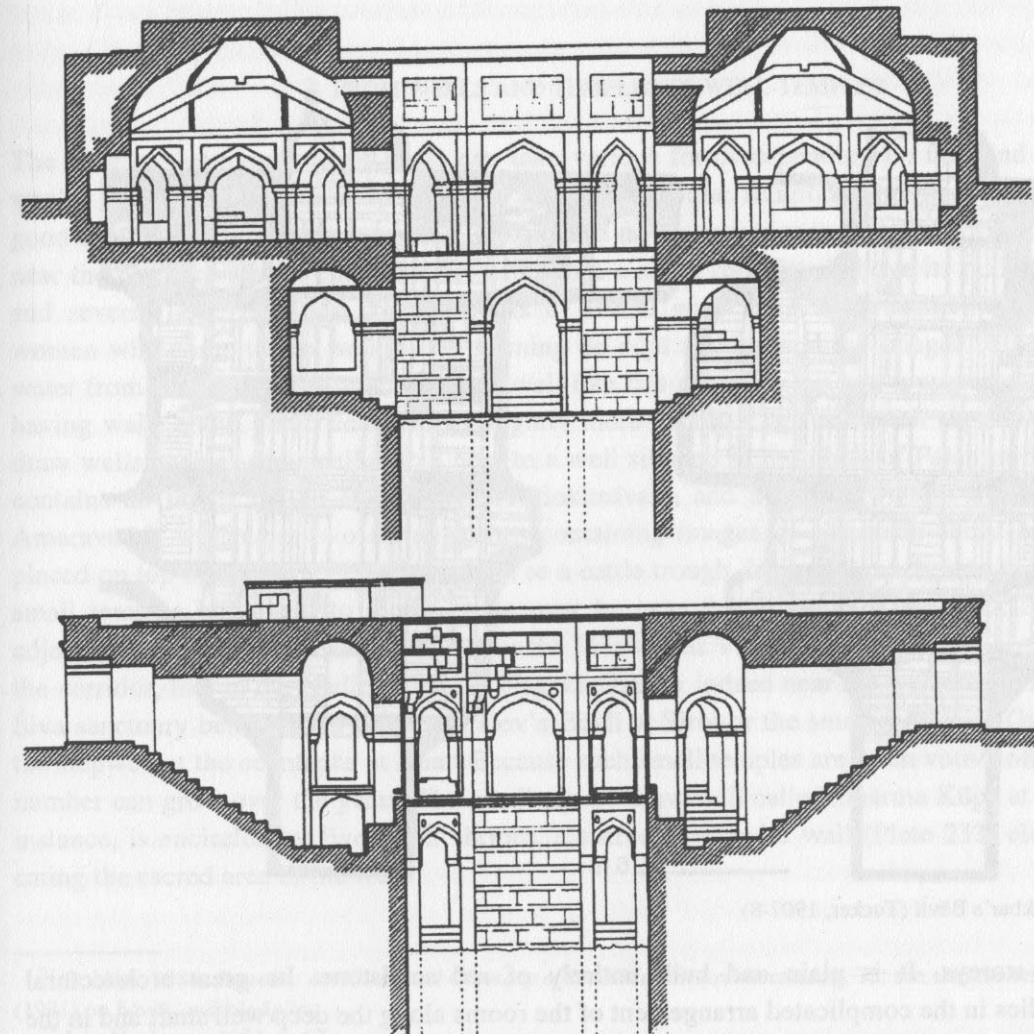


Fig. 34. Fatehpur Sikri: Saṅgīn Bāvlī (Petrucchioli, 1988)

⁶⁸ Jain-Neubauer, 1981, p. 67. For a section and ground plans of the well, see Burgess, 1896, plates 76 and 77.

chambers on top of the *bāvlī* contained treadmills which were turned by men. The water was channelled into a storage tank on the north-west side of the well, from where it was lifted even higher to supply the palace apartments with water. The many wells surrounding the ridge on which Fatehpur Sikri is located, and the remains of complicated water lifting and channelling systems supplying water to the palace buildings all year round, make it very questionable that a shortage of water could have been the reason for the abandonment of the site.⁶⁹ A spiral-well with rooms on three levels and a storage basin beside it is Bābur's Bāvlī in the fort of Agra. Today, the well is built over by the palace structures of the Shāhjahānī Mahal.⁷⁰ Probably the most elaborate example of a spiral-well is located in the south-eastern corner of the fort at Agra attached to the Akbarī Mahal (Fig. 35). The well was constructed by Akbar in the sixteenth century and has six

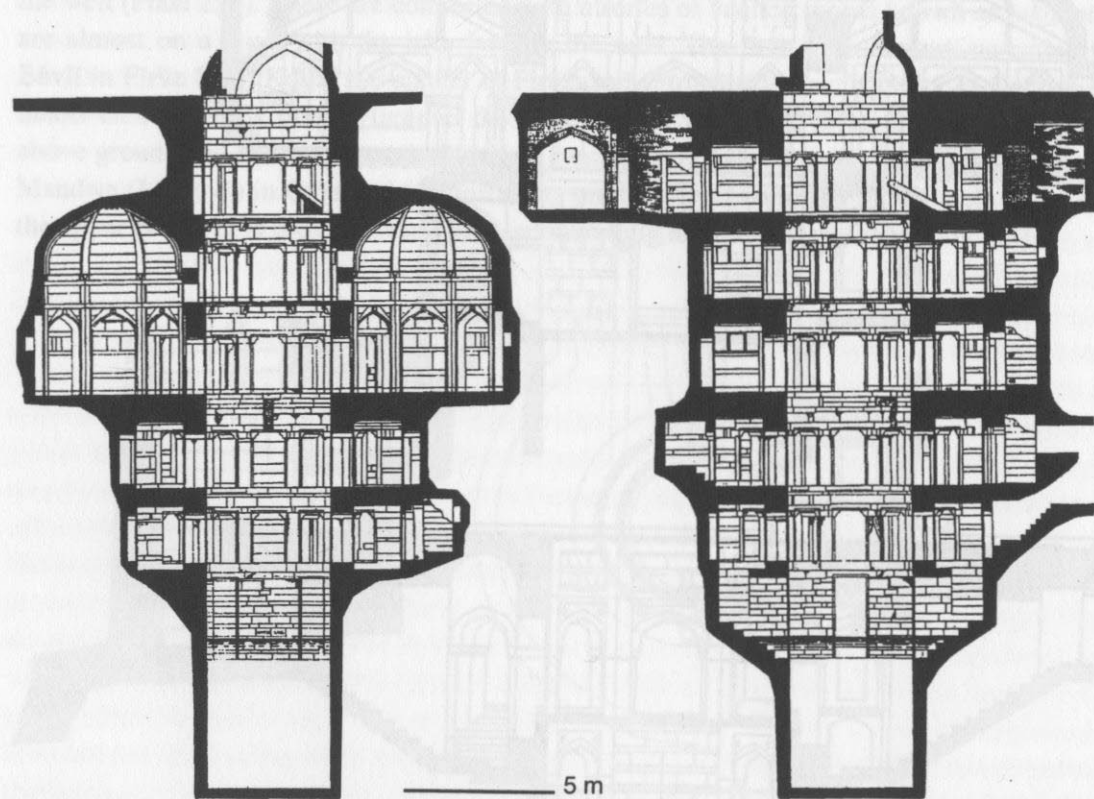


Fig. 35. Agra: Akbar's Bāvlī (Tucker, 1907-8)

underground storeys. It is plain and built entirely of red sandstone. Its great architectural achievement lies in the complicated arrangement of the rooms along the deep well shaft and in the ingenuity of ventilation shafts which supply fresh air even to the lowest storeys of the well. While

⁶⁹ See also Nath, 1985, p. 176.

⁷⁰ Ibid., p. 142.

an arcaded gallery surrounds the first underground storey and leads to rooms and corridors behind, the second and third storeys have large octagonal rooms to the east and the west of the well shaft. The fourth and fifth storeys also have arcaded galleries, and the sixth underground storey has four stairways providing access to the water. Complicated examples such as this are sometimes referred to as *kūpāgāra*, well-houses.⁷¹

A relatively recent example of a spiral-well is to be found in the compound of the summer palace at Wankaner (Wānkāner) in Gujarat, which was constructed between 1930 and 1935 by Mahārāja Rāj Amar Singhji of Wankaner.⁷² The well, which is located under a palace building, has three underground storeys with a large apartment on the second storey and a stepped corridor spiralling around the rectangular well shaft until it meets the water down below.

3. FROM WELLS AND TEMPLES TO WELL-TEMPLES

The religious connotations of wells can take various forms depending on the kind of well, the wealth of the community or the donors, and the particular religious importance of the well. A good example of how little is needed to express a divine presence in a well is a simple draw well near the Guru-kā Mahal⁷³ in Amritsar, which has a small roof fitted above its pulley mechanism and several large mounted popular prints of Hindu gods hung underneath it (Plate 31). The women who come to the well in the morning do *pūjā* and garland the images before they draw water from the well. Over time, a simple well like this may develop into a proper well-temple by having walls and a porch added to it. Slightly more sophisticated are small shrines built next to draw wells such as the small *caitya* next to a well situated in the north of Patan in Nepal, which contains an image of the Bodhisattva Avalokiteśvara, and a simple well in the fields outside Amaravati,⁷⁴ which has two shrine niches containing images of Hanumān and a local goddess placed on top of the parapet wall adjacent to a cattle trough. Many stepwells also have shrines or small temples built next to them. These may be near the entrance to the well, like the ones adjacent to the Ruda Stepwell at Adalaj or the JñānVāv at Visavada; or they may be further along the corridor, like in the Nadir-kī Vāv in Nagaka Gau, or indeed near the well shaft, like the small Śiva sanctuary behind Rāja Bīr Singh Dev's Bāvlī at Sirol or the small temple of Gaṇeśa behind the stepwell at the cenotaphs at Ahar. Because such small temples are often votive offerings, their number can grow over the years. The small square draw well called Dharma Kūpa at Benares, for instance, is encircled by five small shrines, a sacred tree, and a wall (Plate 233) clearly demarcating the sacred area of the well.

⁷¹ R. Nath, 1986, *Jharoka: An Illustrated Glossary of Indo-Muslim Architecture*, p. 73; see also the article by R. Nath (1971) on *bāvlīs* and *kūpāgārs*.

⁷² Jain-Neubauer, 1981, p. 67.

⁷³ Because in a Sikh context *guru* is frequently spelled *gurū*, Gurū-kā Mahal is a further common spelling of this place.

⁷⁴ This is not the place of the famous Buddhist stupa in Andhra Pradesh, but a village already mentioned in the religious introduction for its cow head symbolism, located in Maharashtra.

Among the various sub-groups of well monuments, stepwells are the ones which are usually most elaborately adorned, and many of them resemble underground temple complexes. In addition to small temples built next to them, the side walls of the stepped corridors are often lined with shrines, religious ornamentation and sculptures. Such shrine niches can either be painted with religious symbols, be filled with vegetal ornament, or contain a sculptured religious icon. Vegetal decorations are especially typical of stepwells built under Muslim patronage, as for example the Dādā Harīr Vāv at Asarva. The stepwell at Lakkundi has large four-storeyed shrines on either side, half way down the open stepped corridor. There are a few examples where a shrine room was constructed over the stepped corridor leading down to the well. Examples are the tenth-century Vāpī No. 275 at Sirwal, and the eleventh-century stepwell at Modhera (Plate 219). The tenth-century shrine at the latter site is slightly earlier than the stepwell structure itself and is believed to have been moved to the steps leading down to the well.⁷⁶ In these cases, active worship in an earthly shrine seems to have been offered to the gods in exchange for the violation of their subterranean realm. Towering above the steps of access to the wells, the gods seem to guard the sacred water sites.

It is more typical to find shrines integrated into the back of the well shaft, which is the most sacred part of the well. Although this is less typical of ordinary draw wells, a few examples, such as a well in the north of Patan in Nepal, also have images installed high up on the inside of the well shafts. Looking into the well from above, the images are hardly visible to the person using the well, which are meant more to please and propitiate supernatural forces connected with the water in the well. There is a small *śiva-liṅga* on top of the pulley mechanism of the well adjacent to the Lolārka Kuṇḍa at Benares, whilst the Jñāna-Vāpī Kūpa (Well of Knowledge), also at Benares, is usually covered with a white cloth on the centre of which a *liṅga* made of flower petals is placed. The Jñāna-Vāpī Kūpa is a wide octagonal well contained in an arcade constructed in 1828 by the widow of Mahārāja Daulat Rāo Sindhiyā of Gwalior adjacent to the Viśvanātha Temple in Benares (Plate 25). This well is surrounded by an entire series of myths and stories. One of them recalls a twelve-year drought in the city of Benares and a *ṛṣi* who flung the trident of Śiva into the earth at this place and let the water of the well end the misery of the town. Śiva who heard of the miracle is said subsequently to have taken up his abode in the well.⁷⁷ Other accounts say that when Aurangzeb destroyed the Viśvanātha Temple, either a priest threw the sacred *liṅga* into the well, or that the *liṅga* itself took refuge in a shaft in which subsequently a well, filled by the sacred Ganges, formed to lustrate and protect it.⁷⁸

In stepwells, where one has access to the well shaft from the side, such images and shrines are more easily visible, and often larger and more elaborate. Simple examples where images only were inserted into the upper part of a well shaft can be seen in the stepwell adjacent to the cenotaphs at Ahar, which contains the image of a goddess, allegedly Durgā, and in the stepwell in the

⁷⁵ Vāpī and vāpi both mean well (and also pond). Whilst in some well names a long '-ī' is being used, others spell it with a short '-i'.

⁷⁶ Jain-Neubauer, 1981, p. 57.

⁷⁷ Havell, 1905, p. 182.

⁷⁸ Ibid., p. 182; Gutschow & Michaels, 1993, p. 65.

compound of the Gr̥śneśvara Temple at Ellora, where there is an image of Śiva and Pārvaṭī (Plate 232). The Jethābhāi Stepwell at Isanpur and the Ra Khengar Vāv at Vanthali both have a small shrine integrated into the side of the well shaft further down in the well, while the Ruda Stepwell at Adalaj and the Dādā Harīr Vāv at Asarva have small shrine niches at various levels of the well shaft set one above the other (Plate 234). It is interesting that even in stepwells built under Muslim patronage, the habit of constructing shrines in the shafts of wells continued, although those do not contain divine images but elaborate vegetal decoration, as may be seen in the well shaft shrine of the Dādā Harīr Vāv. In the fourteenth-century stepwell at Khedbrahma (Khedbrahmā) in Gujarat, a horizontal row of small shrines imitating *śikhara* temples runs around the entire inner wall of the well.⁷⁹ Whilst the numerous images lining the inside of the circular well shaft of the Trīkam-Bārot-nī Vāv at Patan are loosely arranged, those adorning the twelfth-century stepwell at Sathambha (Sathāmbha),⁸⁰ both in Gujarat, are set into clear registers. Similarly, in the Rāṇī Vāv at Patan, register above register is filled with hundreds of icons, with Viṣṇu Nārāyaṇa as its main sacred image above the water of the well. An interesting example where an entire temple protruding from the back wall of the well takes up about a third of the well shaft is the Mātā Bhavānī Vāv at Ahmedabad (Plate 237). The shrine which gave the stepwell its name contains an image of the mother goddess Śakti or Ambā Bhavānī⁸¹ and can be reached via a small bridge from the second storey of the last *kūṭa* in front. In the circa nineteenth-century Dhauli Vāv at Ghanerao, this feature seems to have been imitated in the form of a secular balcony overlooking the well (Plate 238). M. Postel described the Mātā Bhavānī Vāv as a plain and mostly utilitarian structure.⁸² With regard to today's actual use of the well, however, this seems most peculiar, since, on the contrary, the well is not used for ordinary water provision but has been completely turned into an underground temple with an ever growing number of new icons, 'well-temple' musicians and active worship. A similar example is a village stepwell in Bhadraka in Gujarat; it has been refurbished with bright colours and tiles, contains many niches with religious icons, and is actively venerated as a temple.⁸³

There are also examples of well-temples where either a shrine or temple was constructed above a well, or where a well was dug into the sanctum of a temple. They date from various periods and are found in different parts of India. A not yet fully evolved example but one clearly going in that direction, is the circa late tenth-century Marīci Kuṇḍa well next to the Mukteśvara Temple at Bhubaneswar (Plate 239). The well, which is located on the south side of the temple, is surrounded by a high wall, and has an entrance gateway with carved pillars and beams resembling those of the ordinary temples of the region. The outer pillars are fashioned in the shape of *nāginīs*,

⁷⁹ The miniature temples contain niches which nowadays no longer house images. There is, however, a local belief which associates this stepwell with the Humad, a group of the local Jain community (Jain-Neubauer, 1981, p. 53).

⁸⁰ This place is also frequently spelled Sāthambā.

⁸¹ Ambā Bhavānī has been identified with Pārvaṭī. Her shrine in the Mātā Bhavānī Vāv seems to be slightly later than the main structure of the stepwell (J. A. S. Burgess, 1905, *Muhammadan Architecture of Ahmadabad*, p. 2. For a plan and section of the well, see plate 13).

⁸² M. Postel in K. Mankodi, 1991, p. 7.

⁸³ Livingston, 1995, pp. 16–7.

half man and half snake, which encircle the doorway with their long twisted tails. The well water is used for temple rituals, and women who cannot have children come to bathe in it. From some distance the well looks like a small dilapidated shrine which has lost its superstructure. A further example of a well-temple resembling a subsidiary shrine is the eleventh-century Bāraj Kuām, also called Tārā Vāv, at Menal (Plate 30). The well is contained within the *vimāna* of the shrine, which had a porch, now almost entirely gone, and steps leading up to the platform bearing the well-temple. The outer walls of the *vimāna* are staggered, decorated with the usual ornaments of the temples of the region, and small shrines which contained images face the cardinal directions. The upper section of the superstructure, of which several columns still remain, seems to have ventilated the well. A larger well monument, called the Mallikār or Manohara Guṇḍam, which resembles the shape of a square *maṇḍapa* with small Nandi sculptures on its roof, is located in the inner enclosure of the Mallikārjuna Temple at Srisailam (circa fourteenth to sixteenth century). It is surrounded by a wall of about 1.2 metres, and contains a large nine metres square basin containing a peculiar construction of columns and beams. The pillars and beams inside the basin and those supporting the roof are all elaborately carved, and there is a standing image of the river goddess Gaṅgā on the west side of the well facing the Mallikārjuna Temple. The pilgrims pay special attention to the finial of the main temple which is mirrored on the water surface of the well. According to the priests of the temple, luck is granted to the person who throws a coin into the well and hits the reflection of the temple finial. This rite stresses the strong link between water structures and temples, and seems to hint at a belief that the god of this temple may at least at times be present in the well. A particularly interesting site for water temples is Bijolia where there are two well-temples on either side of the Mandākinī Kuṇḍa. In the cella of the Bolanāth Mandir to the north-east of the *kuṇḍa* is a well about two and a half metres deep fed by water from the *kuṇḍa*. The *liṅga* contained within the cella is surrounded and often remains submerged by the water of the well. The *liṅga* in the shrine of the Undeśvara Temple to the west is lustrated by a water pot dripping onto it. Its *maṇḍapa* in front has three porches, of which the central one is the entrance, and the one to the south contains a further *liṅga*, also called Bolanāth. The *liṅga* in the main shrine and the Bolanāth-*liṅga* both become submerged by water during the rainy season. The porch to the north contains a spiral staircase leading down to a well which is half inside and half outside the temple wall. A further example of a well-temple is the Pātāleśvara Mandir in the temple group below the Karan Mandir at Amarkantak. The floor of the sanctum is again several metres below ground level and contains a *liṅga* which, once a year on Samanka Pūrṇimā, is submerged when the Ganges emerges here to lustrate the *liṅga*. This belief is also hinted at in the name of the temple, and there are many water structures which are believed to be 'Pātāla Gaṅgās', places where subterranean currents of the Ganges can surface. The Jambukeśvara Temple at Srirangam has a depression in the *garbhagrha* containing the *āp liṅga*, a *svayambhū* or self-created *liṅga* standing in water. A further stage in terms of abstraction is to be found in the *garbhagrha* of the Hātakeśvara Temple at Atri in Orissa. The shrine contains a circular well (about 2.5 metres in diameter) which is encircled by a *yonī* and a metal snake whose head is raised and towers above the well. Whilst no material *liṅga* is present in the waters of this temple, the people worship the well as an inverted water *liṅga* indicated by the *yonī* enclosing it (Plate 235). In well-temples,

water is still venerated in its natural pure form, and has not been substituted for by religious images, such as Gaṅgā Devī or the water god Varuṇa. These wells are clear indicators of the sanctity of water and its identification as a divine substance.⁸⁴

In connection with the idea of a sacred river emerging in a well, it is interesting to note that the source of the Godavari at Brahmagiri above Trimbak, and that of the Son River, outside Amarkantak, are both contained in well-like cavities. The source of the Godavari is a well or cleft hewn into the rock face of the mountain, and the source of the Son is a modern concrete structure bearing the inscription 'Son nadikā udgam', meaning 'source of the Son River'. At Trimbak, the entire mountain seems to be imbued with the sacred river which arises inside it, and whose water can be drawn out of it. At Amarkantak, a few metres away from its source, the Son River is believed to flow into the Narmada, and this underground confluence is believed to be located directly underneath a modern concrete well shaft. Here, the 'well' seems to provide an insight into the underworld, where invisible subterranean rivers merge, and provides the opportunity to draw up the particularly holy water from the sacred confluence.

4. GATEWAYS TO THE UNDERWORLD

Gateways are placed by *ghāṭs* and tanks to mark the transition from the ordinary to the sacred, and at *kuṇḍas* also because they penetrate deep into the ground. All these structures, however, are still open to the sky and flooded with light and air; they are therefore still part of or at least closely connected with our world. Wells penetrate much deeper into the sacred ground; in a stepwell, people themselves descend several storeys under the earth, where it is dark, cool and moist, and where surface life and sounds recede. Imposing gateways and pavilions have been constructed particularly at the entrance stairs to wells in order to mark these subterranean monuments above ground, and to give architectural shape to the transition from our world to the nether regions.

Gateways to wells can be ordinary doorways which make it possible to lock up a well, as may be seen in the eighteenth-century Purāṇa-Ghāṭ-Kanyā Bāvlī at Jaipur; from the outside it has the air of an ordinary entrance in a row of domestic houses. This peaceful well, surrounded by apartments, conveys the feeling of a protected courtyard area away from the busy road. Although today the apartments around the well are inhabited, the well seems originally to have been part of a royal garden. More frequently, however, such doorways are found further down the corridor of a stepwell, as may be seen in the stepwell at the Jaina temples at Khajuraho, or in the Bīja Maṇḍal Bāvlī at Vidisha. The doorway in the latter is found at the first landing, where the two side stairs join the main straight corridor of the well. The doorway is elaborately carved and imitates those leading to shrine rooms of contemporary temples in the region (Plate 236). The remaining pillars at the corners of the landing hint at the presence of an earlier pavilion in front of the doorway. In other cases, *torāṇa* gateways were constructed at the entrances to wells. The well adjacent to the

⁸⁴ There are also temples in which the *garbhagrha* is found on a lower level, or which have two, an upper and a lower *garbhagrha*, separated by a ceiling but where the lower cella remains dry and does not contain water (S. Kramrisch, 1991, *The Hindu Temple*, vol. 1, pp. 171–173; H. Cousens, 1926, *The Architectural Antiquities of Western India*, pp. 38, 53). These examples should not be confused with the well-temples discussed above, in which the water itself or water-*lingas*, visualising the divine presence in the water, are worshipped.

Gaurī Temple at Bhubaneswar is not only marked as sacred ground by a high wall but has a *torana* on its southern side. Other wells have gateways but no walls enclosing them. Stumps of former *toranas* remain for example at the top of the stairs of the Rāñī Vāv at Patan⁸⁵ and at Vāpī No. 1 at Sirwal.⁸⁶ A modern triple *torana* marks the entrance to the sacred compound of the Mātā Bhavānī Vāv, which nowadays is used as a temple. Monumental gateways lead down to the Hāthī Bāvlī outside Adavad and the stepwell in the Kaṭ Golā complex at Murshidabad (Plate 240).

It is typical of Gujarati stepwells to have two gateway towers, which in Gujarati are called *pratoli*, flanking the entrance to wells.⁸⁷ Stone *pratolis* may be seen at the entrance to the Nadir-kī Vāv at Nagaka Gau and the Ruda Vāv at Adalaj; brick-built examples, plastered and painted, can be seen at Uvarsad (Plate 241). Lateral *pratolis* may at times contain shrine niches. Particularly large ones still containing images are present at the Sindhvāi Mātā Stepwell (1633 AD) at Patan in Gujarat. A wall containing an ordinary doorway was inserted between the two *pratolis* flanking the entrance to the Vajināth Vāv at Kapadvanj, and in the Vidyādhara Vāv at Baroda (1487 AD), a domed gateway pavilion was set between the two lateral gate towers (Plate 242). Probably the largest examples of towers on either side of the entrance to a well are those flanking Rājā Bīr Singh Dev's Bāvlī at Sirol (Plate 243).

The majority of stepwells, however, have gateway pavilions above ground clearly marking the entrance and the presence of an underground edifice. Such pavilions can either be domed, as for instance those belonging to the Dādā Harīr Vāv at Asarva (Plate 224), the Jethābhāi Vāv at Isanpur (Plate 216), and the Navalakhi Vāv at Baroda; or they can be pyramidal, as exemplified by the JñānVāv at Visavada (Plate 214), the Vikia Vāv at Ghumli, and the Gaṅgā Vāv at Wadhwan. The Āssi Khāmbhā, a spiral-well in the fort at Gwalior, has a large hall and a gateway resembling the shape of an *īvān* on its northern side (Plate 244), and the sacred Sikh well at Goindwal is entered through a pavilion resembling a miniature gurdwara with a large fluted dome. A particularly interesting example of a gateway to a covered stepwell resembling the shape of a lion is the Simhakiṇaru (Plate 227), in the compound of the Bṛhadiśvara Temple at Gangaikondacholapuram. The people descending to the well seem to pass between the legs of the lion, visualising the fact that the sacred bath in the well is equivalent with one's death and a new birth. The well was constructed by the Zamindar of Udaiyarpalaiyam⁸⁸ who is also believed to have been the donor of two similar lion gateways leading to the water of *kuṇḍas* at Tiruvannamalai, discussed in Chapter Five.

Wells are found all over India, Nepal and Sri Lanka, and continue to be built today. They fall into various sub-groups and have developed features both above and below ground level. Although art-historians have tended to concentrate mainly on stepwell architecture, there are many other

⁸⁵ When Henry Cousens surveyed Patan in 1887, one pillar of the *torana* was still in place (Mankodi, 1991, pp. 30, 44).

⁸⁶ Meister & Dhaky, 1986, p. 156.

⁸⁷ Other current spellings are *pratōlī* and *pratolī*. The use of this term in connection with risalits has been discussed in Chapter Five.

⁸⁸ P. Pichard (ed.), 1994, *Vingt ans après Tanjavur, Gangaikondacholapuram*, p. 149, and fig. 190.

sub-groups of well monuments which show a high degree of elaboration, and artistic and technical ingenuity. All wells penetrate deep into the ground but stepwells and spiral-wells, with their multiple underground storeys and galleried corridors, are in effect large subterranean edifices. Although the decorations of wells are usually related to the local temple architecture, the builders of wells faced entirely different problems from those erecting temples above the ground. The art of building wells evolved greatly over the ages; wells were able to go deeper, have apartments spiralling around the well shaft or adjacent stepped corridors of even greater length and width. While wells in tanks or multiple wells are unusual, there are a few interesting exceptions, such as the Champā Bāvlī at Mandu. This spiral-well is not far from the large Muñj Tālāb, which has further tanks and pools constructed in and next to it. At Kankroli, one of the platforms protruding into the water of Rājasamand Lake contains a deep well fed by the lake, and an example of a multiple well is the Āth-Vāv Nau-Kuām (Eight-wells Nine-wells) at Halol in Gujarat; this has a central octagonal well shaft directly surrounded by sixteen smaller wells.

Because of increased water demand for industry and towns, the ground water table in many regions has dropped considerably, leaving, at least during certain periods of the year, many of the old wells dry. While some wells have been deepened, others have been replaced by modern means of water provision. However, in remote regions and small villages, they remain the predominant means of storing and preserving water all year round.